Daffodil International University
Faculty of Science & Information Technology
Department of Software Engineering
Midterm Examination, Summer 2026
Course Code: SE321; Course Title: Software Engineering Web Application
Section: UC-A; Teacher's Initial: DP

Time: 1 Hour 30 Mins **Marks: 25**

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	***a)***	A customer wants to view a book's details on an online bookstore. The store provides the following link: **https://shop.example.com/books/web/html_basics.html** **Analyze** the anatomy of the given URL and explain how it helps the customer locate the specific book page, rather than only visiting the main domain of the website.	**[Marks-5]**	**CLO- 2** ***Level-4***
	b)	A browser loads a webpage containing **one HTML file and four images** from the same server. The round-trip time (**RTT**) is **80 ms**, and the transmission time is **20 ms per object**. Assume sequential requests, one RTT for each TCP connection setup, and one additional RTT per object request before transmission. Ignore DNS, TLS, and connection closing time. No parallel connections or HTTP pipelining are used. **Calculate** the total loading time for (i) non-persistent HTTP and (ii) persistent HTTP using one reused connection. Show the steps and determine the time saved by using persistent HTTP.	**[Marks-5]**	
2.	***a)***	DIU's online course registration portal allows a student to enroll in an available course. The student opens the portal, selects a course, and clicks the "Enroll" button. The system checks the student's eligibility and available seats, stores the enrollment in a database, and returns a confirmation with an updated course list. **Illustrate** the client-server architecture for this scenario with a labeled diagram. Identify the browser/client, web/application server, and database server. Explain the request-response sequence and state where the user interface, validation/business logic, and persistent data storage are handled.	**[Marks-5]**	**CLO- 3** ***Level-3***

	b)	**Construct** an HTML membership form for a university sports club using a table layout. The form should include fields for Member Name, Student ID, and Age. There should also be Preferred Sports using checkboxes. Finally, the form must contain both a Submit button and a Reset button to allow users to submit the form or clear the entered information. The code Output should be: **Sports Club Membership Form** **Member Name** [ ] **Student ID** [ ] **Age** [ ] **Preferred Sports** ☐ **Football** ☐ **Cricket** ☐ **Badminton** **Submit** **Reset**	**[Marks-5]**	**CLO- 3** ***Level-3***

3.	***a)***	You are developing a small password validation system for a website. Each user creates a password during registration. **Develop a JavaScript program using a regular expression and basic DOM manipulation that:** 1. Takes a password from an HTML input field. 2. Validates the password when the user clicks a button. 3. Displays the result on the webpage. Conditions: • The password must contain at least 8 characters. • It must include at least one uppercase letter, one lowercase letter, one digit, and one special character from @#$!%*?&. • No whitespace is allowed. • If all conditions are met, show: "Valid Password"; otherwise, show: "Password Unacceptable". Sample Output Example - Input: abc Output: Password Unacceptable	**[Marks-5]**	**CLO- 4** ***Level-6***